PLASMA WAVES

General Treatment

we are working with an infinite and uniform Plasma. For waves in a linear medium:

$$\nabla \times \vec{B} = \mu_0 j + \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} \quad ; \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Solution must have variables that go like: $\exp i(\vec{k}\cdot\vec{x} - \omega t)$

Fourier analysis yields: $i\vec{k} \times \vec{B} = \mu_0 \hat{j} + \frac{-i\omega}{c^2}\vec{E} \quad ; \quad i\vec{k} \times \vec{E} = i\omega \vec{B}$

Some eq^n manipulation: $\vec{k} \times (\vec{k} \times \vec{E}) + \frac{\omega^2}{c^2}\vec{E} + i\omega\mu_0 \vec{j} = 0$

Need to solve plasma eq^n's to get this.

we can use the generalized ohm's law: $\vec{j} = \vec{\sigma}\cdot\vec{E}$

Substituting: $\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E} + \frac{\omega^2}{c^2}\vec{E} + i\omega\mu_0\vec{\sigma}\cdot\vec{E} = 0$

Our plasma can be treated as a dielectric material driven by current density $\vec{j}$ instead of polarization $\vec{P}$. The ampere's law will be:

$$\frac{1}{\mu_0}\nabla \times \vec{D} = \vec{j}_{ext} + \frac{\partial \vec{D}}{\partial t} = \frac{\partial}{\partial t}\epsilon\epsilon_0\vec{E}$$

Our generalized ohms law.

Subbing maxwell's eq^n:

$$\vec{j} + \epsilon_0\frac{\partial \vec{E}}{\partial t} = \vec{\sigma}\cdot\vec{E} + \epsilon\frac{\partial \vec{E}}{\partial t} = \frac{\partial}{\partial t}\epsilon_0\vec{E}\cdot\vec{E} \longrightarrow \vec{\epsilon} = 1 + \frac{\vec{\sigma}}{-i\omega\epsilon_0} = 1 + i\frac{\mu_0 c^2}{\omega}\vec{\sigma}$$

This is our plasma's dielectric tensor. This allows to manipulate our first eq^n:

$$k(k\cdot\vec{E}) = -k^2\vec{E} + \frac{\omega^2}{c^2}\vec{\epsilon}\cdot\vec{E} = 0 \longrightarrow \vec{D}\cdot\vec{E} = 0 \quad \text{where} \quad \vec{D} = \left\{\vec{k}\vec{k} - k^2\vec{I} + \frac{\omega^2}{c^2}\vec{\epsilon}\right\}$$

To have a non-zero $\vec{E}$, we require: $\det|\vec{D}| = 0$ which will give us an eq^n that relates $\vec{k}$ and ω, which is our dispersion relation.

An isotropic medium can have two sol^n:

① $-k^2 + \frac{\omega^2}{c^2}\epsilon = 0 \longrightarrow E_z = 0$ we have a transverse field; $\omega/k = c/\sqrt{\epsilon}$

This behaves like a regular EM wave w RI: $\sqrt{\epsilon}$.

② $\epsilon = 0 \longrightarrow E \parallel \vec{k}$ This is a longitudinal field and has similar properties as a dielectric

② $\epsilon = 0 \rightarrow E \parallel k$ This is a longitudinal field and longitudinal wave which is permitted in plasma physics.

Generally:

$$D = \frac{\omega^2}{c^2}\begin{pmatrix} -M^2+\epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & -M^2+\epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{pmatrix} \; ; \; M^2 = \frac{k^2c^2}{\omega^2}$$

A determinant will give us a $\pm M^2$. Two modes of a wave. Polarization will be generally longitudinal partly and partly transverse.

The above discussion applies to all linear mediums. For plasmas, we need to solve for $\vec{\sigma}$.

High Frequency Plasma Conductivity

We will be operating under the following assumptions:

$\nabla p = 0 \rightarrow$ Cold Plasma. $\vec{v}_0 = 0 \rightarrow$ Stationary Eq^m.

$\bar{\nu}_{ei} = 0 \rightarrow$ Collisionless.

For $\vec{B}_0 = 0$; Plasma isotropic mom^n eq^n:

$$mn\left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right] = nq\vec{E} \longrightarrow m\frac{\partial \vec{v}}{\partial t} = q\vec{E} \rightarrow \vec{v} = \frac{q}{-i\omega m}\vec{E} \rightarrow \vec{j} = nq\vec{v} = \frac{nq^2}{-i\omega m}\vec{E}$$

This gives us: $\epsilon = 1 - \left(\frac{nq^2}{m\epsilon_0}\frac{1}{\omega^2}\right) = 1 + \chi$

For longitudinal waves, $\epsilon = 0 \rightarrow \omega_p = \left(\frac{n_e e^2}{m_e \epsilon_0}\right)^{1/2}$ This is the electron plasma frequency.

Longitudinal oscillations of both ions and electrons in an unmagnetized plasma occur at one plasma (e$^-$/ion) frequency.

Electrons can alternatively be modelled as simple harmonic oscillators at plasma frequency.

Interestingly, note now the wave's group velocity v_g is 0:

$v_g = \frac{d\omega}{dk} = 0 \rightarrow$ There is no energy/information travel from these waves.

Transverse wave dispersion is different though, and is given by:

$\omega = (\omega_p^2 + k^2c^2)^{1/2}$

When $M^2 < 0$, whats happening?

Here M is purely imaginary and the wave is "Evanescent", which means it just cuts

off / damps exponentially.

If wave is incident on a plasma with $\omega_p^2 > \omega^2$, it is simply reflected.